GRADUATION PROJECT PROPOSAL

Two Candidate Projects in Analog / Mixed-Signal and Digital ASIC Design

Idea A: AetherRV droop-aware RISC-V AI chip Idea B: 400 Gb/s ADC-based serial-link receiver

DOCUMENT	Project proposal for review and selection
DISCIPLINE	Electronics and Communications Engineering (ASIC / AMS design)
TEAM SIZE	7 members per project (analog and digital)
DATE	October 2026

1. Purpose of this document

This proposal presents two candidate graduation projects so that the supervisors and any industry partner can evaluate them side by side. Both projects are mixed-signal in nature: in each, an **analog block is the key technical challenge** and a digital block turns it into a complete, verifiable system. Each idea is described with its problem, proposed solution, evaluation method, work plan, team and risks. A direct comparison and the requested next steps are given on the last page.

2. The two ideas at a glance

IDEA A | AetherRV

A RISC-V chip with two CPUs and an INT8 AI accelerator, protected by an **analog supply-voltage monitor** that detects droops. A controller then throttles accelerator activity and requests safe clock changes.

CORE ANALOG IDEA
Voltage-droop detector
TARGET
22nm FD-SOI implementation
PROOF
4-mode comparison on one AI workload

IDEA B | 400 Gb/s ADC-Based Receiver

A serial-link receiver that digitizes the incoming signal with an ADC and recovers the data with digital signal processing. Current links run at about 200 to 224 Gb/s; this project aims at **400 Gb/s**.

CORE ANALOG IDEA
High-speed ADC front end
TARGET
Digital-on-top mixed-signal chip
PROOF
SER / BER versus an IEEE reference paper

3. Comparison summary

	Idea A: AetherRV	Idea B: 400 Gb/s ADC-based receiver
Problem	Supply droop can cause timing failures in AI hardware	Channel loss limits data rate; 400 Gb/s is beyond current 200 to 224 Gb/s links
Analog role	Voltage monitor and droop detector (core)	ADC and analog front end (core)
Digital role	RISC-V CPUs, AI accelerator, droop controller	Equalization, timing recovery and detection DSP
Current status	Architecture, roles and 10-month plan defined	MATLAB 224 Gb/s model running; SER already better than the reference paper

Details, risks and a full comparison table follow in the next pages.

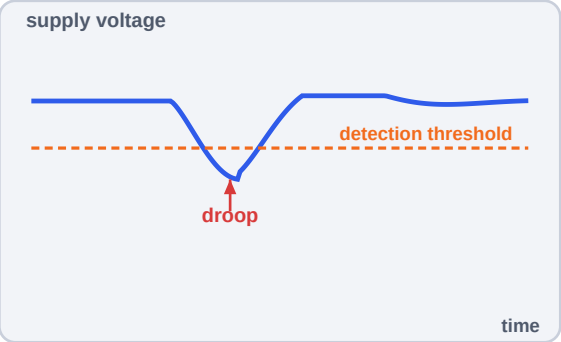
Idea A: AetherRV

A RISC-V chip that runs AI and responds to supply-voltage dips detected by an analog monitor

A.1 Background and problem statement

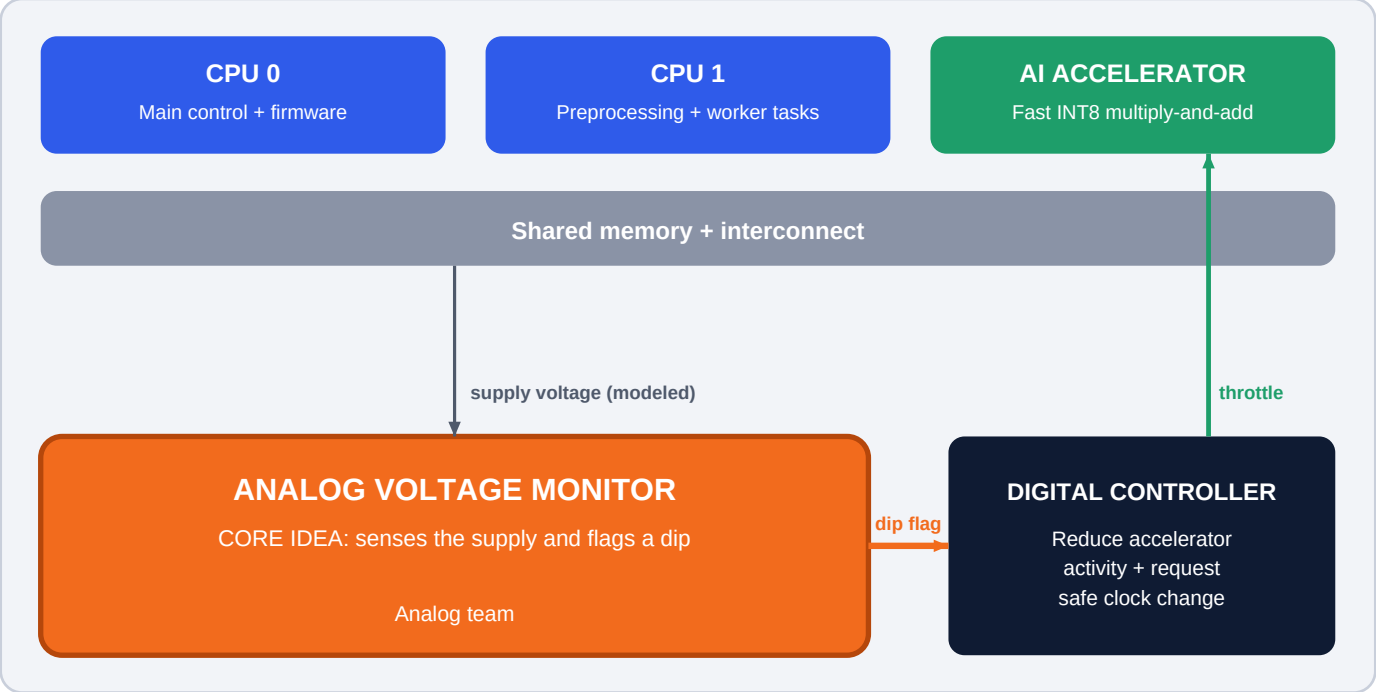
When many circuits switch at the same time, they draw a sudden surge of current. This briefly pulls the supply voltage down, an effect called a **voltage droop**. A lower supply voltage makes circuits slower, so operations may miss their timing limits and produce wrong results. An AI accelerator performing massive multiply-and-add work in parallel is exactly the kind of load that triggers droops.

Problem statement: how can a chip running AI workloads detect a droop fast enough, and respond in a way that keeps timing safe without wasting too much speed or energy?



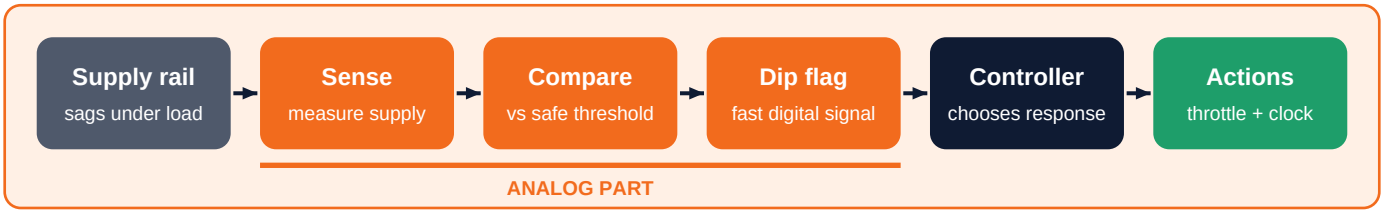
A.2 Proposed solution and chip architecture

The design combines two RISC-V CPUs, an INT8 AI accelerator, shared memory and an interconnect, and adds a **supply-voltage monitor feeding a droop controller**. The monitor measures the supply and raises a dip flag; the controller reduces accelerator activity and requests safe clock changes, then lets the system recover and resume.



A.3 The analog part is the core idea

A voltage droop is an **analog event**: the supply physically sags and recovers within a very short time. The CPUs, accelerator and controller can only react if something **measures the sag first**. The analog monitor is that component, and the rest of the system exists to use what it sees. Without it the chip is blind to droop.



Idea A: AetherRV (continued)

Objectives, evaluation, work plan, team and risks

A.4 Objectives and novelty

- Design and verify a dual-RISC-V SoC with an INT8 AI accelerator, shared memory and interconnect.
- Build a voltage-droop detector (behavioral model first, with a custom analog detector as an extension) and a controller closing the droop-response loop.
- Coordinate **activity throttling and clock control** using both the voltage measurement and the AI workload activity.
- Quantify the trade-off between energy, speed and droop severity against fixed, clock-only and activity-only baselines.

Novelty: most responses rely on the voltage reading alone. Combining it with workload activity is the research contribution. *This still needs literature review and experimental validation.*

A.5 Evaluation methodology

The same AI inference workload is run in four modes under the same supply model and operating conditions:

A Fixed baseline	B Clock-only	C Activity-only	D Combined AetherRV
No adaptive mitigation	Lower clock on a dip	Limit accelerator rate	Coordinate activity + clock

Measured: prediction correctness, peak voltage dip, energy per inference, execution time, and added hardware area and power.

Initial goals (not achieved results): at least 10% lower peak droop and 5% lower energy per inference, with no more than 10% added latency, relative to the matched baselines defined in the plan.

A.6 Work plan (10 months)

Months	Phase	Main activities
1 - 3	Build the foundation	Architecture, interfaces, first hardware modules
4 - 6	Integrate and control	Run AI on the chip and close the droop-control loop
7 - 8	Implement and check	Physical design, timing and power analysis
9 - 10	Compare and present	Experiments, thesis and final demonstration

A.7 Team and roles

Role	Members
Analog (voltage monitor and detector)	Fahd, Kareem Amr
Digital ASIC	Sakr, Nourhan
Digital verification	Islam, Mohamed Ali
MATLAB, system architecture and ASIC	Basel

A.8 Deliverables, scope and risks

Core deliverables: verified RTL, a behavioral supply and detector model, reproducible AI benchmarks, and available physical-design results. **Optional extensions:** a custom transistor-level analog detector and fabricated silicon.

Risk	Mitigation
Novelty not confirmed in literature	Early literature review; fall back to a rigorous comparison study
Targets (10% / 5% / 10%) not met	Report measured results honestly against matched baselines
Timing closure or physical-design delays	Fix interfaces early; keep a lean architecture

Idea B: 400 Gb/s ADC-Based Serial-Link Receiver

A receiver that digitizes the link signal with an ADC and recovers data using digital signal processing

B.1 Background and motivation

Serial links carry data between chips, boards and servers, and the demand for bandwidth keeps pushing the data rate of each link upward. Current high-speed links operate at about **200 to 224 Gb/s**. At such rates the channel strongly attenuates and distorts the signal, causing inter-symbol interference. An **ADC-based receiver** converts the received waveform to digital samples early, so that powerful equalization and detection can be done in the digital domain, which is flexible and scales well with modern technology.

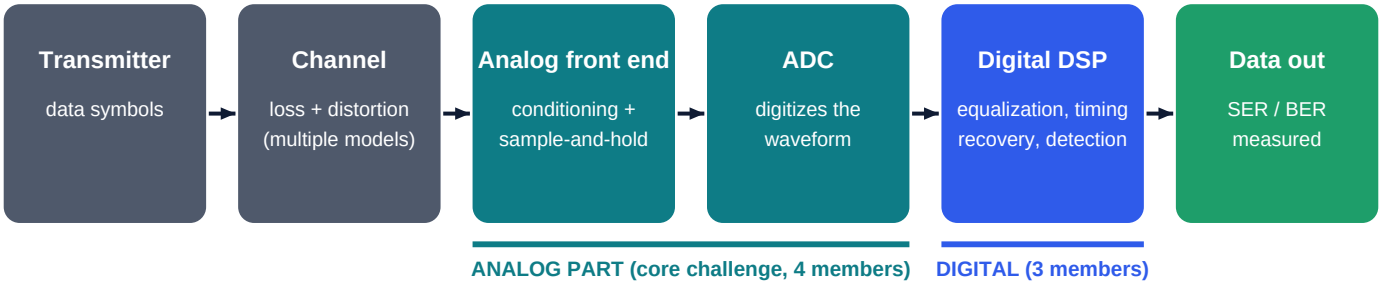
B.2 Problem statement and objectives

Problem statement: can an ADC-based receiver architecture be designed to reach **400 Gb/s** while keeping the symbol error rate (SER) and bit error rate (BER) acceptable over realistic channels?

- Reproduce and then beat the SER / BER reported in a reference IEEE paper for a 224 Gb/s ADC-based link, using MATLAB.
- Evaluate the design across multiple channel models, not a single channel.
- Extend the architecture and algorithms toward 400 Gb/s.
- Implement the digital receiver blocks in RTL and design the key analog blocks, targeting a digital-on-top chip.

B.3 Proposed receiver architecture

Planned architecture, to be refined during modeling.



MATLAB end-to-end model: used to explore the architecture and compare SER / BER with the reference paper

B.4 Progress to date

- An IEEE reference paper on 224 Gb/s ADC-based serial links was identified as the benchmark.
- A MATLAB model of the 224 Gb/s link has been built; the achieved **SER is already better than the paper's**.
- BER improvement is still in progress, and the model is being tested over more than one channel.
- The MATLAB code repository is available for review on request.
- Academic guidance is provided by Dr. Abdelghani; the digital members already know the ASIC flow.

Why this is a strong AMS project: the ADC and analog front end determine how much of the channel distortion can be corrected digitally. The system-level trade-off between analog performance and digital equalization effort is the heart of modern high-speed link design.

Idea B (continued) and Comparison

Work plan, team, risks, side-by-side comparison and requested next steps

B.5 Proposed work plan (10 months, to be confirmed)

Months	Phase	Main activities
1 - 2	Reference and baseline	Literature review; reproduce the reference paper in MATLAB at 224 Gb/s
3 - 4	Architecture exploration	Reduce BER; test multiple channels; explore the path to 400 Gb/s; derive analog block specifications
5 - 7	Block design	Analog front end and ADC design; digital DSP RTL and fixed-point model
8 - 9	Integration and verification	Mixed-signal co-simulation, RTL verification, digital-on-top physical flow
10	Results and presentation	Final comparison, thesis and demonstration

B.6 Team

Group	Size
Analog design	4 members
Digital design (knows ASIC flow)	3 members
Academic guidance	Dr. Abdelghani
Contact	Ziad Yasser Mohamed (Communications, Batch 28)

B.7 Deliverables

- MATLAB end-to-end model with SER / BER results across several channels
- Architecture and specification document
- Verified RTL for the digital receiver
- Analog block designs with simulation results
- Target: digital-on-top chip layout, subject to resources

B.8 Risks

Risk	Mitigation
400 Gb/s is beyond current published links	Tiered goals: 224 Gb/s beating the paper first, then 400 Gb/s exploration
ADC speed, resolution and power trade-offs	Specify analog blocks from the MATLAB model before circuit design
BER gap versus the SER already achieved	Dedicated algorithm work and multi-channel testing in months 3 - 4

9. Side-by-side comparison

Criterion	Idea A: AetherRV	Idea B: 400 Gb/s receiver
Core analog block	Supply-voltage droop detector	ADC and analog front end
Digital content	CPUs, AI accelerator, controller	Equalization and detection DSP
Key evidence	4-mode comparison of energy, speed, droop	SER / BER versus IEEE reference paper
Current status	Plan, architecture and roles defined	MATLAB model running, SER better than paper
Main risk	Novelty and targets unproven	400 Gb/s is very ambitious

10. Requested next steps

- Review both ideas and give feedback on feasibility, scope and technology availability.
- Advise which idea (or which tier of an idea) best fits an industry-supported graduation project.
- Indicate whether the MATLAB repository and the team CV should be shared for a detailed review.

GRADUATION PROJECT PROPOSAL | IDEA C

Supply-Aware Energy-Proportional PAM-N Wireline Receiver

Reconfigurable multi-stage Decision Feedback Equalization (DFE) coordinated with power-integrity-aware control

Purpose. This third candidate builds directly on the team's existing 224 Gb/s MATLAB receiver model and connects it to research themes in reconfigurable wireline equalizers, energy-proportional operation, and power-management ICs.

CORE DIGITAL IDEA	CORE ANALOG / POWER IDEA
Reconfigurable multi-stage, multi-tap PAM-N DFE with selectable speculation/configuration modes.	Supply-disturbance and regulator behavioral models; evaluate one feasible monitor or power-management circuit block.
TARGET	**PRIMARY PROOF**
Verified RTL and synthesis/timing/area evaluation; 224 Gb/s baseline first, 400 Gb/s as a stretch goal.	Matched comparison of BER/SER, energy per bit, and recovery under supply disturbances and channel variation.

C.1 Background and research question

High-speed PAM-N wireline receivers trade equalization capability against digital complexity and energy. A reconfigurable multi-stage DFE can provide different operating points, while supply disturbances and regulator behavior can affect receiver performance and power. This project will investigate whether coordinating DFE configuration with supply-aware operating decisions can meet a target error rate with lower energy than fixed or independently adaptive alternatives.

Research question: Can a joint DFE-configuration and supply-aware control policy reduce energy per recovered bit while maintaining a specified BER/SER under realistic channel and supply-disturbance conditions?

C.2 Proposed architecture

PAM-N transmitter + channel model
Analog front end / ADC behavioral model
Reconfigurable multi-stage DFE + symbol detector
BER/SER and energy estimator
Configuration controller using channel quality + supply status + error target
Supply disturbance / LDO model provides modeled supply status and operating constraints

Scope principle: begin with behavioral models and a validated digital baseline. Implement transistor-level power-management or analog circuitry only after the supervisor confirms feasibility, tool access, and schedule.

C.3 Objectives and proposed research contribution

1. Reproduce the existing 224 Gb/s MATLAB receiver baseline and document SER/BER under matched channel conditions.
2. Implement a reconfigurable PAM-N DFE with defined tap/stage/speculation modes and quantify their complexity.
3. Build a behavioral supply-disturbance/regulator model and model its impact on selected receiver parameters.
4. Develop a controller that selects a valid DFE operating point using channel quality, supply status, and an error-rate constraint.
5. Compare joint adaptation against fixed DFE, DFE-only adaptation, and supply-aware control without DFE adaptation.
6. Evaluate the path toward 400 Gb/s after the baseline and contribution are validated.

Potential novelty (to be confirmed by literature review)

The proposed novelty is not adaptation alone. It is the measured coordination of reconfigurable DFE complexity and supply-aware operating decisions under explicit BER/SER and energy constraints. The literature review must determine whether this exact combination is new and how it differs from prior energy-proportional DFE and power-aware receiver work.

C.4 Evaluation methodology

MODE	DFE CONFIGURATION	SUPPLY CONTROL	PURPOSE
A	Fixed	Fixed operating point	Reference baseline
B	Adaptive DFE	Fixed operating point	Isolate DFE adaptation
C	Fixed DFE	Supply-aware	Isolate supply-aware control
D	Jointly adaptive	Supply-aware	Evaluate proposed method

Metrics: BER, SER, energy per bit, DFE complexity, synthesized area, timing, and recovery behavior under supply disturbances. Include channel variation and matched input data; report measured results without presuming improvement.

C.5 Seven-person team structure

ROLE	COUNT	OWNERSHIP
Digital ASIC design	2	Reconfigurable DFE RTL; synthesis, timing, area and implementation flow.
Digital verification	2	Self-checking DFE tests, MATLAB/RTL equivalence, controller and regression verification.
Analog / power	2	Supply/regulator behavioral models, receiver analog impairment modeling, one feasible circuit block if practical.
Project leader (Basel)	1	System architecture, adaptation algorithm, integration, research coordination and digital implementation.

IDEA C | WORK PLAN, DELIVERABLES AND RISKS

C.6 Ten-month work plan

MONTHS	PHASE	MAIN ACTIVITIES
1–2	Baseline and research gap	Reproduce 224 Gb/s MATLAB results; literature review; define BER/SER and energy targets.
3–4	Reconfigurable DFE	Implement selectable DFE modes; quantify complexity; create supply/regulator behavioral model.
5–6	RTL and controller	Implement DFE RTL and adaptation controller; build verification environment; compare MATLAB and RTL.
7–8	Integration and evaluation	Run four-mode matched experiments; synthesize RTL; evaluate timing, area and energy proxies.
9–10	Research results	Consolidate results, assess 400 Gb/s stretch path, complete thesis and prepare paper/demo.

C.7 Deliverables

- Reproducible MATLAB baseline and channel/supply test scenarios.
- Reconfigurable DFE RTL and self-checking verification environment.
- Behavioral supply/regulator and receiver-impairment models.
- Matched four-mode BER/SER, energy, complexity, area, and timing comparisons.
- Synthesis/physical-design results available from the tool flow.
- Thesis and a paper-ready evaluation; 400 Gb/s extension and custom analog implementation are stretch goals.

C.8 Main risks and mitigation

RISK	MITIGATION
Novelty overlap with prior DFE/power-aware work	Conduct an early focused literature review; define a precise gap with the supervisor before committing.
Supply effect on receiver not sufficiently modeled	Use a documented behavioral model first; validate assumptions and sensitivity; implement only supported coupling mechanisms.
400 Gb/s scope too ambitious	Treat 224 Gb/s reproduction and improvement as the core milestone; 400 Gb/s is conditional.
Analog circuit schedule risk	Keep behavioral models as core deliverables; select one circuit block for transistor-level design only if feasible.
Power claims unsupported by RTL alone	Separate measured simulation/synthesis results from estimates; use consistent activity and power assumptions.

C.9 Questions for supervisor review

1. Does the joint DFE-configuration and supply-aware optimization offer a defensible research gap?
2. Which DFE configurations and complexity metrics should be prioritized?
3. Which supply-disturbance coupling model is physically meaningful and feasible with the available tools?
4. Which circuit block, if any, should be implemented at transistor level?
5. What should be the minimum publishable result, and what should remain a stretch goal?

Status: proposed concept for supervisor discussion. Novelty, circuit feasibility, and target improvements remain to be validated.